

HCK4G

CTRL-ALT-REPAIR:

HACKING THE RIGHT TO FIX

BY

DACK SESSIONS

&

DAVID MULRANEY

ROOT://_WHO ARE WE



SAFETY FIRST NO EXPLOSIONS, PLEASE

WEAR GLOVES. ALWAYS.

USE PROTECTIVE EYEWEAR (ESPECIALLY NEAR SCREENS).

DON'T STAB THE BATTERY.

DON'T HEAT DEVICES UNNECESSARILY.

LITHIUM-ION = MINI BOMB IF PUNCTURED WHILE CHARGED.

ASSUME THE BATTERY IS ALWAYS LIVE.

NO FOOD/DRINK NEAR WORKSTATIONS.

SAFETY CONTINUED HAZARDS TO EXPECT

⚡ SHORT CIRCUITS (ESPECIALLY ON EXPOSED BOARDS)

🔥 BURN RISK FROM SOLDERING IRONS

💥 BATTERY EXPANSION OR VENTING

💧 BROKEN GLASS THUS DISPOSE CAREFULLY, DON'T TOUCH DIRECTLY



EQUIPMENT OVERVIEW

- 34 iPhones (various models)
- 28 with screens (for teardown + demo)
- 6 headless demo phones (for board access)
- 4 soldering stations + heat guns, spudgers, isopropyl, gloves, screw mats

iPhone hoe: Ew u have an android.
Ur so poor.

Meanwhile their 5 year old iPhone 6:



WORKSHOP OBJECTIVE

- Learn iPhone teardown workflow (screen-off to NAND)
- Identify chip layout and critical forensic points
- Discuss chip-off risks and limitations
- Practice safe board handling



TEARDOWN WORKFLOW OVERVIEW

POWER CHECK (MULTIMETER)

HEAT SCREEN → PRY OPEN

BATTERY DISCONNECT FIRST!

BOARD EXPOSURE

SHIELD REMOVAL

CHIP IDENTIFICATION

WHAT WE'RE LOOKING FOR

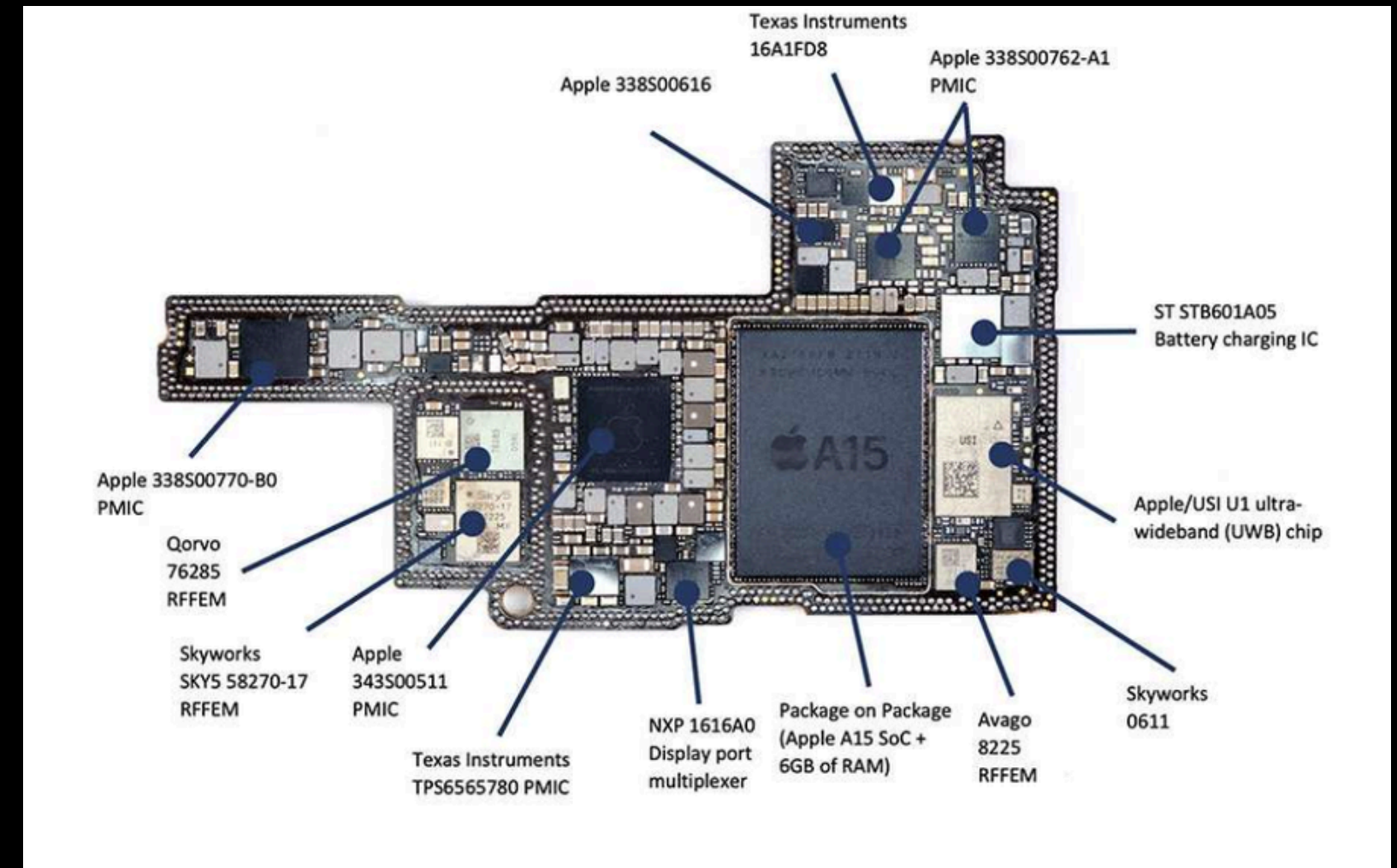
NAND FLASH (DATA STORAGE)

CPU / SECURE ENCLAVE

BASEBAND CHIP

EEPROM / SMALL LOGIC CHIPS (E.G. WIFI/BLEETOOTH)

PHYSICAL INDICATORS OF TAMPER / WATER / DAMAGE



HANDS ON PROTOCOL

Form small groups: 1 phone per 1-2 people

- **You'll have 20-1 hour to:**
- **Safely disassemble**
- **Identify major chips**
- **Practice reassembly (screen optional)**

Rotating through screenless demo phones for closer inspection if needed and please ask questions!!! :)

BONUS TOPICS (TIME PERMITTING)

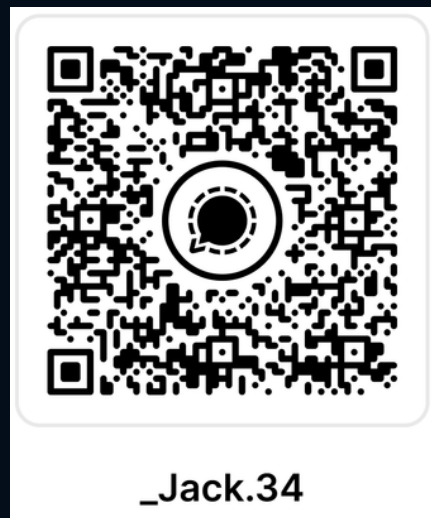
NAND removal basics (no reballing today)

Passive data extraction overview

JTAG / ISP vs Chip-Off

Why Secure Enclave ruins everyone's day 😅

THANK YOU



WWW.LINKEDIN.COM/IN/DACKSESSIONS



DACKSESSIONS@PROTONMAIL.COM



WWW.LINKEDIN.COM/IN/DAVID_MULRANEY



DAVID@THEREPAIRGROUP.COM.AU